

# **GASTROINTESTINAL PHARMACOLOGY**

# **Drugs for Peptic Ulcer Disease (PUD) and Gastro- Esophageal Reflex Disease (GERD)**

# Peptic Ulcer Disease (PUD)

- PUD- characterized by ulcer formation in the esophagus, stomach, or duodenum areas of the GI mucosa that are exposed to gastric acid and pepsin
- PUD can be;
  - Gastric
  - Duodenal
- Causes;
  - Stress
  - Drugs (NSAIDs)
  - H. pylori
- Pathophysiology;
  - Imbalance between cell destructive and cell protective effects



# Pathogenesis;

## Aggressive Factors

H. pylori

Drugs (NSAIDs)

Acid

bile

Pepsin



## Protective Factors

Mucus

Bicarbonate

Blood flow

Cell renewal

Prostaglandins

Somatostatin



# Parietal Cell & Acid Regulation

❖ Gastric acid secretion from parietal cells is induced by three factors:

➤ **Ach** from vagus nerve

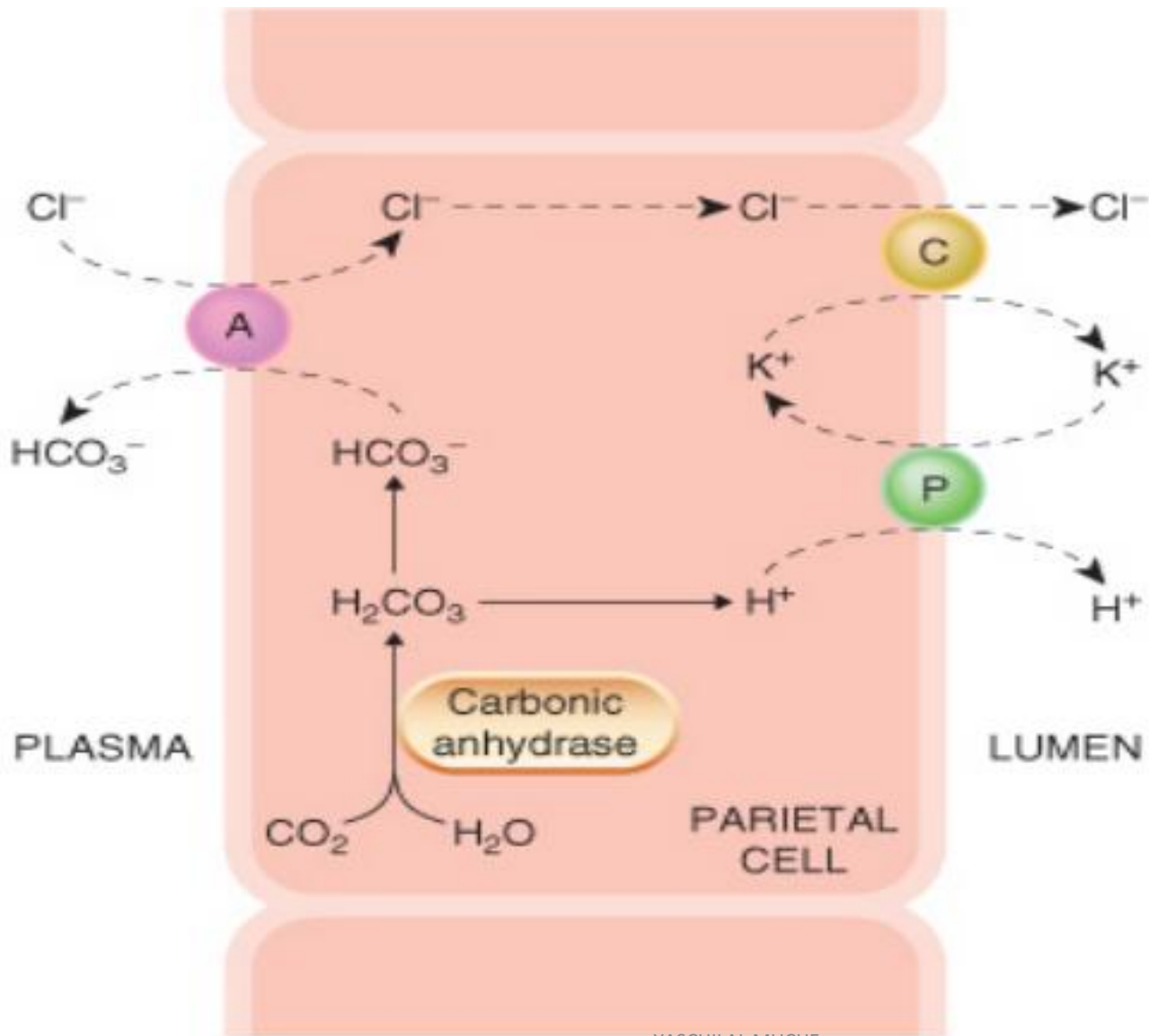
➤ **gastrin** from antral G cells in response to food ingestion and stretching of the stomach wall.

➤ **histamine** from Enterochromaffin like cells in gastric mucosa

❖ Parietal cells contain  $H^+-K^+$  ATPase which is responsible for exchange of  $H^+$  and  $K^+$ .

➤  $H^+$  is combine with  $Cl^-$  ions in the gastric lumen to form HCL

# Parietal Cell Acid Secretion



# Gastro esophageal Reflux Disease (GERD)

- Backflow of stomach acid into the esophagus
- Main symptoms are irritation and burning in chest or throat.
- More severe symptoms: difficulty in swallowing, chest pain
- In some patients (~10%), the normal esophageal lining or epithelium may be replaced with abnormal (Barrett's) epithelium.
  - This condition (Barrett's esophagus) has been linked to esophageal cancer

# Drugs for PUD and GERD:

- ☐ Antacids
- ☐ H<sub>2</sub> Receptor Blockers
- ☐ Proton Pump Inhibitors
- ☐ Prostaglandin Analogs
- ☐ Mucosal Protective Agents
- ☐ Anti-cholinergics (eg; Pirenzepine)
- ☐ Eradicating the H. pylori infection with anti-microbial agents

# Antacids

- Antacids are Weak bases in nature
  - **$\text{Al}(\text{OH})_3$ ,  $\text{Mg}(\text{OH})_2$ , Magnesium tri silicate**
- React with HCl in the stomach
  - Produce less acidic and poorly absorbed salts
  - Raise the PH of gastric secretions
  - indirectly inhibiting activity of pepsin (pepsin is inactive above pH=4)
- Commonly used antacids are combinations of Al and Mg compounds.

# *H<sub>2</sub>-receptor antagonists*

- ✓ *Cimetidine, Ranitidine, Nizatidine and Famotidine.*
- ✓ *competitively block the binding of histamine to H<sub>2</sub> receptors,*
  - *reduce secretion of gastric acid.*
- In usual doses, all **inhibit 60–70% of total 24-hour acid secretion.**
- **Effective at inhibiting nocturnal acid secretion**
  - **which depends largely on histamine**

## **Uses;**

- PUD
- Gastroesophageal reflux disease (GERD)
- Zollinger-Ellison syndrome (a gastrin-producing tumor causes increased HCl secretion)

# Proton Pump Inhibitors (PPIs)

- Omeprazole, esomeprazole, lansoprazole, dexlansoprazole, rabeprazole, and pantoprazole.
- MOA: are irreversible, direct inhibitors of the proton pump ( $K^+/H^+$  antiport) in the gastric parietal cell.
- inhibit 90–98% of 24-hour acid secretion
- inhibit both fasting and meal stimulated secretion

## Uses:

- PUD, GERD and Zollinger–Ellison syndrome.
- Included in eradication regimen for *H. pyloric* induced PUD.
- Px and Tx of NSAID induced ulcer

# Misoprostol

- A congener of prostaglandin E1
  - stimulate mucus and bicarbonate secretion
  - enhance mucosal blood flow.
  - reduce histamine-stimulated acid secretion.
- Contraindicated during pregnancy
  - induce abortion, premature birth or birth defects.
- Most common AEs that limit its use
  - Dose-related diarrhea (10 to 40%) and nausea

# Mucosal Protective Agents

- Agents that enhance mucosal defense
- Also known as cytoprotective compounds

## 1) Sucralfate

- Forms sticky polymer in acidic environment and adheres to the ulcer site → forming a protective barrier

## 2) Chelated Bismuth (Bismuth subsalicylate)

- antimicrobial action,
- inhibits the activity of pepsin,
- increases secretion of mucus, and
- interacts with glycoproteins in necrotic mucosal tissue to **coat and protect the ulcer**

# Therapy of *H. pylori*-induced PUD

## Two drug regimens for eradication of *H. pylori*:

### ❖ Triple Therapy - 14 day treatment - Effective 80-85%

- Proton pump inhibitor + amoxicillin (metronidazole in penicillin allergy) + clarithromycin

### ❖ Quadruple Therapy - 7 day treatment (PPI based) = as efficacious as triple therapy

- Bismuth salicylate + metronidazole + tetracycline + either a PPI or H2RA

# **Antidiarrheal drugs**

- Diarrhea; Increased GI tract motility and decreased absorption of fluid
- Antidiarrheal drugs include;
  - ✓ ant motility agents,
  - ✓ adsorbents,
  - ✓ drugs that modify fluid and electrolyte transport

## Antimotility agents

- **diphenoxylate** and **loperamide**
- MOA: activate presynaptic opioid receptors in the enteric nervous system → inhibit acetylcholine release → decrease peristalsis

## Adsorbents

- **aluminum hydroxide** and **methylcellulose**
- MOA: **adsorbing intestinal toxins or microorganisms** and/or coating or protecting the intestinal mucosa.
- much less effective than antimotility agents
- interfere with the absorption of other drugs

## Agents that modify fluid and electrolyte transport;

- Bismuth subsalicylate

- ✓ **MOA; decreases** fluid secretion in the bowel
- ✓ used for traveler's diarrhea
- ✓ Adverse effects; black tongue and black stools.

# **LAXATIVES (Drugs for constipation)**

- Classified on the basis of their mechanism of action:
  - Irritants and stimulants (**Senna, Bisacodyl, Castor oil**)
  - Bulk laxatives (**methycellulose, psyllium seeds, and bran**)
  - Saline and osmotic laxatives (***magnesium citrate and magnesium hydroxide***)
  - Stool softeners (**docusate sodium and docusate calcium**)
    - ✓ become emulsified with the stool → produce softer feces and ease passage
  - Lubricant laxatives (**Mineral oil and glycerin suppositories**)
    - ✓ act by facilitating the passage of hard stools
  - Chloride channel activators (***Lubiprostone***)